JKLU

LAPTOP PRICE PREDICTOR

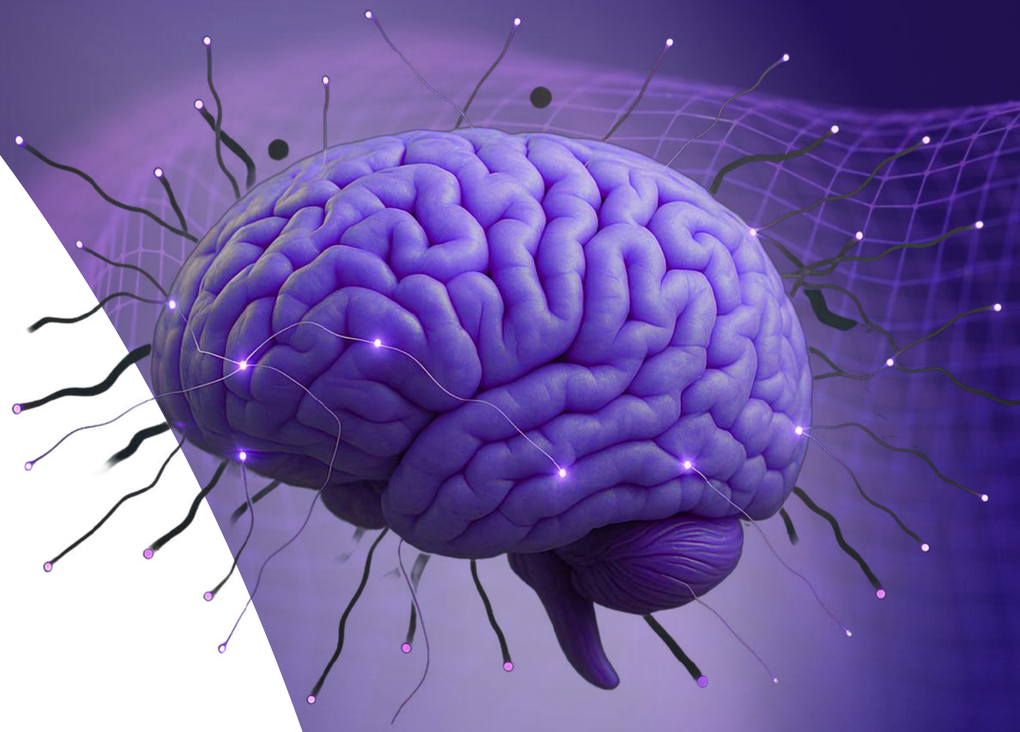
MACHINE LEARNING PROJECT

Presented By:

Apoorv Sharma (2024BTech258)
Lakshya Vashishtha (2024BTech072)
Rohit Soni (2024BTech013)

PROBLEM STATEMENT

The increasing variety of laptop brands and configurations has made it difficult for consumers to estimate the correct price of a laptop based on its specifications. Existing pricing methods rely on manual comparison or static listings, which do not accurately reflect the combined impact of hardware and display features on pricing. This project aims to develop a machine learning–based laptop price prediction system that accurately estimates the selling price using key specifications such as brand, processor, RAM, storage, GPU, and display features, helping users make informed and cost-effective purchasing decisions.



DATASET

Name : Laptop Price Predictor
Dataset taken from Kaggle

Dataset link : https://www.kaggle.com/datasets/anubhavgoyal10/laptop-prices-dataset

LITERATURE REVIEW

BY Lakshya Vashishtha

Paper Reference	Chosen Dataset Parameters (Input X / Output y)	Research Paper Parameters (Input X / Output y)	Difference Between Both	Extra Parameters Added / Not Added & Reason
Karakuş & Bilgin (2024) – Laptop Price Range Prediction with ML	**Input (X):** Brand, Laptop Type, Ram (in GB), Weight, OS, GPU, Touchscreen, IPS Display, Hard Drive, SSD, Screen Resolution, Screen size, Processor Name **Output (y)**: Laptop Price	**Input (X):** Hardware specifications, storage types, display attributes, processor details after preprocessing and feature engineering **Output (y):** Laptop Price Range (Low, Medium, High)	The research focuses on classifying laptops into price ranges instead of predicting exact prices and uses heavy preprocessing and boosting algorithms, while our project directly predicts actual price values using core specifications.	No additional parameters were added because exact price prediction is more practical for real-world usage and the core hardware features already capture major price influence.
Singh et al. (2025) – Laptop Price Estimation Using Predictive Analytics	**Input (X):** Brand, Laptop Type, Ram (in GB), Weight, OS, GPU, Touchscreen, IPS Display, Hard Drive, SSD, Screen Resolution, Screen size, Processor Name **Output (y)**: Laptop Price	**Input (X):** Brand, CPU, GPU, RAM, storage type, display features, OS, touchscreen, IPS panel **Output (y):** Laptop Price	The research uses extensive data cleaning, feature engineering, and multiple ML algorithms to optimize performance, while our project uses a simpler pipeline focused on essential laptop features.	No extra features were included since both datasets already contain the most influential hardware attributes affecting laptop pricing.
Tian (2024) – Predictive Model Using Linear Regression, RF & XGBoost	**Input (X):** Brand, Laptop Type, Ram (in GB), Weight, OS, GPU, Touchscreen, IPS Display, Hard Drive, SSD, Screen Resolution, Screen size, Processor Name **Output (y)**: Laptop Price	**Input (X):** Brand, Screen Size & Resolution, CPU, GPU, RAM, Storage (HDD/SSD), OS, Weight **Output (y):** Laptop Price	The paper emphasizes model comparison and feature importance analysis across regression and ensemble models, whereas our project prioritizes building a practical and interpretable price prediction system.	No additional parameters were added as the same fundamental hardware features sufficiently explain laptop price variation.
Ghosal et al. (2024) – ML-Based Laptop Price Prediction Across Brands	**Input (X):** Brand, Laptop Type, Ram (in GB), Weight, OS, GPU, Touchscreen, IPS Display, Hard Drive, SSD, Screen Resolution, Screen size, Processor Name **Output (y)**: Laptop Price	**Input (X):** Brand, CPU, RAM, Storage type, GPU, Display features, Weight, OS, Touchscreen, IPS **Output (y):** Laptop Price	The research integrates AutoML tuning, feature engineering and multiple ensemble models, while our project focuses on direct supervised learning using structured laptop specifications.	No extra parameters added because both models already incorporate core performance and display features essential for accurate pricing.
Raj et al. (2023) – Intelligent Laptop Trading Application with Price Prediction	**Input (X):** Brand, Laptop Type, Ram (in GB), Weight, OS, GPU, Touchscreen, IPS Display, Hard Drive, SSD, Screen Resolution, Screen size, Processor Name **Output (y)**: Laptop Price	**Input (X):** Laptop specifications, historical pricing data, market trends, brand, storage, processor, GPU **Output (y):** Predicted Laptop Price	The paper combines price prediction with an e-commerce trading system using market trend data, while our project strictly focuses on technical specification-based price estimation.	Market trend and economic indicators were not included to maintain dataset simplicity and focus on hardware-driven pricing.

LITERATURE REVIEW

BY **ROHIT SONI**

Paper Reference	Chosen Dataset Parameters (Input X / Output y)	Research Paper Parameters (Input X / Output y)	Difference Between Both	Extra Parameters: Added / Not Added & Reason
LAPTOP PRICE PREDICTION, International Journal of Novel Research and Development	**Input (X):** Brand, Laptop Type, Ram(in GB), Weight, OS, GPU, Touchscreen, IPS Display, Hard Drive, SSD, Screen Resolution, Screen size, Processor Name **Output (y):** Laptop Price	**Input (X):** Laptop Brand, Laptop Model / Type, Processor (CPU), RAM (in GB), Storage Type (HDD / SSD), Storage Capacity (in GB), Graphics Card (GPU), Display Type (IPS / Non-IPS), Touch Screen (Yes / No), Operating System **Output (y):** Laptop Price	Both datasets use largely similar configuration-based parameters; differences arise mainly from feature granularity and naming conventions	**Not added.** Existing features sufficiently represent the laptop price.
Analysis of Laptop Price, School of Wu Xiang Middle School	**Input (X):** Brand, Laptop Type, Ram, Weight, OS, GPU, Touchscreen, IPS Display, Hard Drive, SSD, Screen Resolution, Screen size, Processor Name **Output (y):** Laptop Price	**Input (X):** Laptop ID, Brand (Company), Laptop Type (TypeName), Screen Size (Inches), Screen Resolution, Processor (CPU), RAM (GB), Storage / Memory, Graphics Card (GPU), Operating System (OpSys), Weight (kg)**Output (y):** Laptop Price	The research paper includes additional detailed hardware attributes, while the implemented dataset focuses mainly on core, interpretable pricing features.	**Not added.**Extra technical features increase model complexity without significant improvement in price prediction.
Laptop Price Prediction, Department of Computer Technology, KDKCE & RTMNU University	**Input (X):** Brand, Laptop Type, Ram, Weight, OS, GPU, Touchscreen, IPS Display, Hard Drive, SSD, Screen Resolution, Screen size, Processor Name**Output (y):** Laptop Price	**Input (X):** Laptop Model / Type, RAM, ROM (HDD / SSD), Graphics Card (GPU), Processor (CPU), IPS Display, Touch Screen **Output (y):** Laptop Price	The first dataset includes additional system-level and physical attributes, while the research paper focuses on core configuration features directly influencing laptop price.	**Not added.**Additional attributes increase model complexity without providing significant improvement in price prediction accuracy.
Laptop Price Prediction, Dantuluri Narayana Raju College, Bhimavaram, Andharapradesh	**Input (X):** Brand, Laptop Type, Ram, Weight, OS, GPU, Touchscreen, IPS Display, Hard Drive, SSD, Screen Resolution, Screen size, Processor Name**Output (y):** Laptop Price	**Input (X):** Laptop Company, Product Type, Screen Size, Weight, RAM, CPU Speed, Price **Output (y):** Laptop Price	The chosen dataset includes more detailed hardware and system-level features, whereas the research paper uses a smaller set of core attributes focused on basic laptop specifications.	**Not added.** Additional features increase model complexity and are not essential for effective laptop price prediction..
Analysis of Machine Learning, Bina Nusantara University, Jakarta, Indonesia	**Input (X):** Brand, Laptop Type, Ram, Weight, OS, GPU, Touchscreen, IPS Display, Hard Drive, SSD, Screen Resolution, Screen size, Processor Name**Output (y):** Laptop Price	**Input (X):** Brand, Processor Speed, RAM Size, Storage Capacity, Screen Size, Weight **Output (y):** Laptop Price	The chosen dataset contains more detailed hardware and display-level features, while the research paper uses a reduced set of core numerical specifications focused on price prediction.	**Not added.** Additional features increase dimensionality and model complexity without proportional improvement in prediction accuracy.

LITERATURE REVIEW

BY Apoorv Sharma

Paper Reference	Chosen Dataset Parameters (Input X / Output y)	Research Paper Parameters (Input X / Output y)	Difference Between Both	Extra Parameters: Added / Not Added & Reason
Real-Time Laptop Price Prediction through Web Data Analysis, G. Anudeep Goud et al., ICCSCE, 2025	**Input (X):** Brand, Laptop Type, RAM, Weight, OS, GPU, Touchscreen, IPS Display, Hard Drive, SSD, Screen Resolution, Screen Size, Processor Name**Output (y):** Selling Price	**Input (X):** Brand, Processor Type, RAM, Storage (HDD/SSD), GPU, Laptop Specifications (real-time scraped)**Output (y):** Laptop Price	Paper uses real-time web-scraped data and dynamic price tracking, whereas our dataset is static and specification-based with additional display and physical features.	**No extra parameters added** because real-time scraping and price trend history are beyond the scope of an academic ML project and core pricing features are already covered.
Laptop Price Estimation Using Machine Learning, Puvvada Pragnatha et al., IJRESM, 2024	**Input (X):** Brand, Laptop Type, RAM, Weight, OS, GPU, Touchscreen, IPS Display, Hard Drive, SSD, Screen Resolution, Screen Size, Processor Name**Output (y):** Selling Price	**Input (X):** Company, RAM, Type Name, Screen Size, Screen Resolution, Memory, OS, GPU, CPU, Weight**Output (y):** Laptop Price	Research paper relies on static datasets with extensive feature engineering, while our dataset directly includes user-oriented hardware and display features for simpler modeling.	**No extra parameters added** as engineered features (CPU grouping, log transforms) are model-specific and not mandatory at dataset level.
Webscraping Laptop Prices to Estimate Hedonic Models, Zafar & Himpens, 2019	**Input (X):** Brand, Laptop Type, RAM, Weight, OS, GPU, Touchscreen, IPS Display, Hard Drive, SSD, Screen Resolution, Screen Size, Processor Name**Output (y):** Selling Price	**Input (X):** Brand, RAM, Storage, CPU frequency, Cores, GPU type, Screen specs, Dimensions, OS, Touchscreen**Output (y):** Laptop Price (Hedonic estimation)	Paper focuses on hedonic pricing models and economic price indices, whereas our project focuses on direct ML-based price prediction.	**No extra parameters added** because hedonic and economic features increase complexity and are unnecessary for a student-level ML project.
Laptop Price Range Prediction with Machine Learning Methods, Karakuş & Bilgin, IJMSIT, 2024	**Input (X):** Brand, Laptop Type, RAM, Weight, OS, GPU, Touchscreen, IPS Display, Hard Drive, SSD, Screen Resolution, Screen Size, Processor Name**Output (y):** Selling Price	**Input (X):** Company, Type, Screen Size, Screen Resolution, CPU, RAM, Memory, GPU, OS, Weight, Touchscreen, CPU Speed**Output (y):** Price Range (Class)	Paper treats the problem as a classification task (price ranges), while our project is a regression task predicting exact price.	**No extra parameters added** because price-range labels and classification-specific features are not required for regression-based prediction.
Laptop Price Prediction using Machine Learning Algorithms, M. A. Shaik et al., IEEE ICETEMS, 2022	**Input (X):** Brand, Laptop Type, RAM, Weight, OS, GPU, Touchscreen, IPS Display, Hard Drive, SSD, Screen Resolution, Screen Size, Processor Name**Output (y):** Selling Price	**Input (X):** Company, Laptop Type, OS, Processor, RAM, Memory, Screen Resolution (PPI), Touchscreen, Weight**Output (y):** Laptop Price	Research paper performs heavy preprocessing like PPI calculation and memory splitting, whereas our dataset keeps explicit hardware attributes for simplicity.	**No extra parameters added** as engineered features increase preprocessing complexity without significant benefit for academic prediction accuracy.

Reference

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THANK YOU

Hope you get the best prices for your laptops after using our model

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